

My current hypothesis is based on the letter configurations in the ENDYARH portion of K3. The letters Y, A, and R are raised ever so slightly above the rest of other letters in the first line of K3 (or superscripted). Upon closer inspection, D and Y as well as R and H seem like they're paired together, while A is just hanging out there by itself. My first question was, "Why would Sanborn cut those letters into the copper sheet (which isn't thin) without first double checking to ensure they were properly aligned with the rest of the letters? Was he drunk when he did that? Why would he purposely superscript those letters for a piece of art that he knew was going to last for a long time? Why would he accept those imperfections in his final product?" But the more I think about it, I believe the superscripting was done on purpose.

My next thought was, "Ok. If those letters were superscripted on purpose, what was that purpose?" I thought back to the scenario Sanborn was trying to convey through the artwork. If the sculpture was supposed to be a message shared between a handler and his spy, where face-to-face communication was next to impossible, how would the handler get a message to his spy (or vice versa) without getting caught? My mind goes to the Soviet Union with the KGB breathing down everyone's neck and making human intelligence operations next to impossible. To me, the person writing the message would have to communicate the decryption method on the first encounter, because there was no way to assume a second meeting would occur. If that's the case, where would one look to find that decryption method? Here is where I think sections K1, K2, and K3 of KRYPTOS come further into play.

If you look at the decrypted letters of K1, you'll notice that the word IQLUSION is situated all way to the right-hand side of the message. Moving down to K2, you'll notice that UNDERGRUUND is also on the right-hand side of the message, but not as far to the right as IQLUSION. And if you move down to K3, DESPARATLY is on the center-left of the message. If you draw a diagonal line that intersects these decrypted words, you get something that *might* be indicating where to look for the method of decrypting K4, which just so happens to be the seven beginning letters of K3, ENDYARH.

So, what would the superscripted letters mean if they are to be viewed as a method of decryption? One way to interpret the D/Y and the R/H letters in ENDYARH is that one of those letters is the beginning and ending points for the first row/column of the transposition. Perhaps we need to start the first transposition at the letter R and go in a way that ends the first row/column of transposed letters with H, then follow that with the second row/column of transposed letters beginning with the letter after H until the second row/column of transposed letters matches the number of letters in the first row/column, and so on until the letters run out.

Another interpretation could be that we should work with both the encrypted and decrypted letters of K3 to begin and end our transposition. Perhaps we start with O, which is the decrypted letter that corresponds to D in ENDYARH. It's the "O" in

SLOWLY. After that, we end our first line of transposed letters with the encrypted letters, so Y. There's only one letter "Y" in K4, so it's possible that it could be an indicator or letter of importance when doing the first transposition. If we were to layer the decrypted message immediately underneath or on top of the encrypted message, we would be going from one layer to another when we begin and end our first line of the transposition.

Another possibility is that the first letter of the transposition (D or R) into a row/column 1 needs to be placed in such a way that it mirrors the superscripting of the letters in K3 so that Y or H of row/column 2 are diagonally located from the first letters. I'm not entirely sure which way we're supposed to transpose the letters, so this is a complete guess on my part. But I do believe transposing the letters is supposed to be the first step in solving K4.

What makes me think the superscripted A is intended to mean a substitution or Vigenere cipher should be applied after doing the first transposition? Well, for me, it's just weird to have one letter hanging about by itself when the two other superscripted letters seem to be just a tad bit closer to the other letters in K3. If the letter A is supposed to indicate a substitution should be carried out, does that mean only one letter should be used as the substitution method? Possibly. This would be like a Caesar shift where every letter is changed by the same key letter in the message's alphabet. If this is the case, which letter should be used as the substitution key? I've already gone through every single letter shift in the key alphabet to find that the letters Y, T, S, B, I, and J are the only letters that, when applied to the original encoded message, decrypt to the minimum number of letters needed to have EASTNORTHEAST and BERLINCLOCK in the resulting message.

Another possibility is that a key schedule, or set of letters that can be applied to the entire message, is applied to the transposed letters to get an entirely new set of letters. This would be a multi-word Vigenere cipher and the decrypted letter counts would depend on the initial transposed letter's position. If we are supposed to use the previous sections to help solve K4, it's possible that we would need to use a key schedule that included IQLUSION, UNDERGRUUND, and DESPARATLY. It would mean KRYPTOS is a sequential discovery puzzle, where solving one puzzle leads to the solving of another. This would be very similar to the puzzle processes that are seen in many escape rooms. The only potential key schedule that uses the previously misspelled words in the decrypted sections is:

DESPARATLYUNDERGRUUNDIQLUSION. If repeated three times and DESPARATLY is added to the end, the exact number of letters in the key schedule is 97. I previously wrote about this potential key schedule in message #22625.

What we're left is six possible substitution letters and one key schedule, for a total of seven possibilities that could be used to change the letters of the first transposition of

letters.

To me, it would make sense to assume that if one of the superscripted letters in K3 suggest transposition in the beginning, then the other superscripted letter should be treated the same. But how to get to that point after first changing the positions of the letters and then the letters themselves is a challenge.

One of the reasons I believe the decryption method is supposed involve a transposition, followed by a substitution, and ending with another transposition is that this method is mentioned in the Army's technical training manual on Basic Cryptography. The section is called Monoalphabetic and Polyalphabetic Substitution Combined with Transposition and can be found on page 154. It's a form of super encryption, which I'm pretty sure we can assume Sanborn used in KRYPTOS, given its difficulty. It's a way to apply what was done to solve sections K1-K3, but combining the methods and changing the order.

I'm still working on this approach, but am definitely willing to see if anyone has any ideas on how to tweak it.